

Program 7: Deploy a web application to kubernetes using minikube

File structure:

Nginx-deployment.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

 name: nginx-deployment

spec:

 replicas: 2

 selector:

 matchLabels:

 app: nginx

 template:

 metadata:

 labels:

 app: nginx

 spec:

 containers:

 - name: nginx

 image: nginx:latest

 ports:

 - containerPort: 80

Start the minikube service and check the status

 #minikube start

 #minikube status

Create the deployment

 #kubectl apply -f nginx-deployment.yaml

Check the deployment status

 #kubectl get deployments

 #kubectl get pods

Expose the port

 #kubectl expose deployment nginx-deployment --type=NodePort --port=80

Get the service name

 #kubectl get services

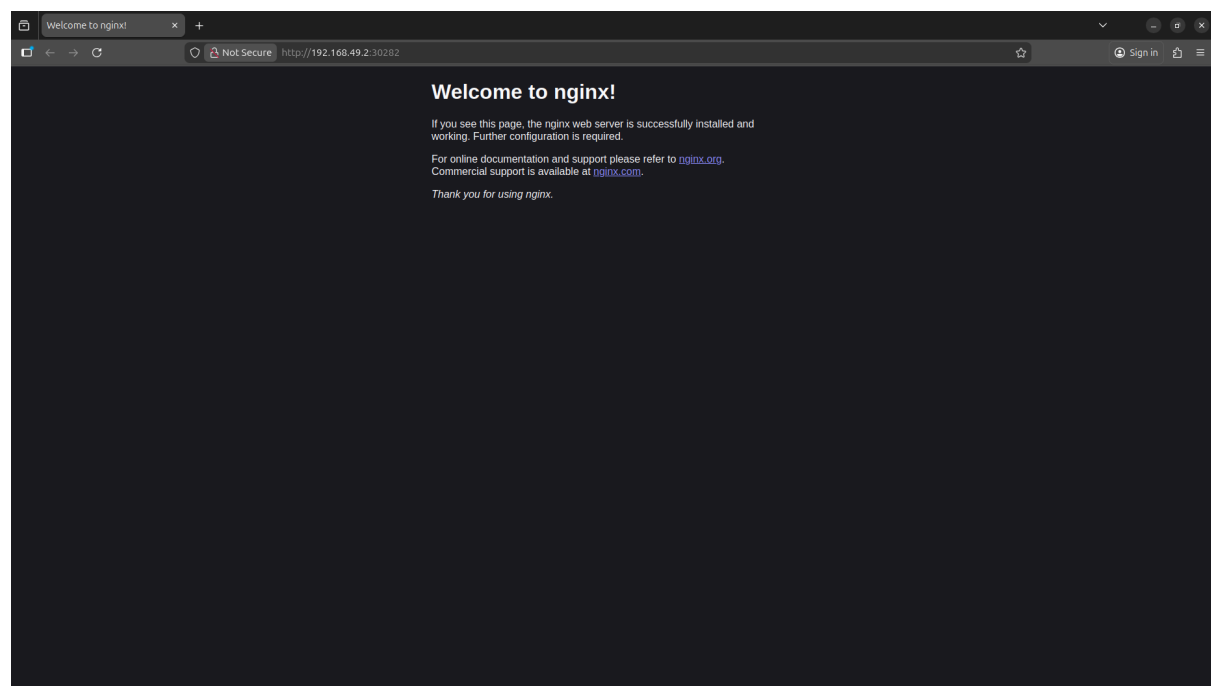
```
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/Lab7
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms$ cd Lab7
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
🔥 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.40 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner

! /usr/bin/kubectl is version 1.30.14, which may have incompatibilities with Kubernetes 1.34.0.
   ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$ code nginx-deployment.yaml
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  1/2     2            1           12s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-6r75n  1/1     Running   0           23s
nginx-deployment-6f9664446b-shbrl  1/1     Running   0           23s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$ kubectl get services
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP     10.96.0.1     <none>         443/TCP           140m
nginx-deployment  NodePort      10.105.60.249 <none>         80:30282/TCP     10s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/Lab7$
```

Start the service

#minikube service nginx-deployment



Deleting resources

#kubectl delete pods --all

#kubectl delete svc--all

#kubectl delete deployments --all